

Water Resources: Planning, Hydrology, and Sustainability

Hydrological Cycle and Global Water Distribution

Water resources encompass all natural supplies of water in liquid, solid, or vapor states. Freshwater accounts for only approximately 2.5 percent of the Earth's total water volume, with the remaining 97.5 percent residing in ocean saltwater. Of the total freshwater, over 68 percent is trapped in ice caps and glaciers, approximately 30 percent exists as groundwater, and less than 1 percent is readily accessible in rivers, lakes, reservoirs, and soil moisture.

The global hydrological cycle drives the continuous movement of water through evaporation, transpiration, condensation, precipitation, infiltration, and runoff. Human activities such as deforestation, urban development, land conversion, and climate variability significantly alter precipitation patterns, river streamflow regimes, and basin evaporation rates, impacting overall water availability.

Integrated Water Resources Management (IWRM)

Integrated Water Resources Management (IWRM) is an internationally accepted operational framework designed to coordinate the development, management, and allocation of land, water, and related ecological resources. The principal objective of IWRM is to maximize economic prosperity and social welfare without compromising the sustainability of vital ecosystems.

IWRM incorporates cross-sectoral decision-making across agricultural, industrial, domestic, and environmental water sectors. Key principles include basin-level river planning, economic valuation of water services, stakeholder participation, gender equality in governance, and transparent regulatory mechanisms to handle competing allocation demands during periods of peak scarcity.

River Basin Hydrology and Reservoir Operations

A river basin or watershed represents the fundamental geographic unit for hydrologic analysis and water management. Modern basin management utilizes remote sensing, satellite radar altimetry, GIS mapping, and rainfall-runoff hydraulic modeling to track streamflow, sediment transport, and river channel dynamics.

Multipurpose reservoirs provide storage capacity for flood mitigation, hydroelectric power generation, agricultural irrigation, and municipal drinking water supply. Reservoir rule curves dictate seasonal drawdown and filling cycles based on hydrologic forecasts, ensuring multi-year carryover storage while preserving environmental flow requirements for downstream aquatic habitats.

Climate Change Adaptation and Water Security

Climate change accelerates hydrologic extreme events, shifting seasonal snowpack melt timelines, expanding arid climate zones, and increasing the frequency of heavy precipitation downpours. Water security requires adaptive planning capable of absorbing hydrological shocks without systemic failure.

Resilience measures combine structural interventions (such as expanding storage, inter-basin transfers, and flood detention basins) with non-structural strategies (including demand management, water metering, rainwater harvesting, watershed reforestation, and water trading markets).